

Emma Jane Weller

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Education

Yale University – 2024-present

Ph.D. Candidate in Astronomy

Thesis Advisor: Priyamvada Natarajan

Harvard University – 2020-2024

A.B. in Physics and Astrophysics

Summa Cum Laude, GPA: 4.00

Fellowships & Awards

- ★ **Sigma Xi Honor Society** – 2026
- ★ **National Science Foundation Graduate Research Fellowship** – 2024
- ★ **Sterling Prize Fellowship** (*Yale University*) – 2024
- ★ **Summa Cum Laude** (*Harvard University*) – 2024
- ★ **Highest Honors in Physics** (*Harvard University*) – 2024
- ★ **Phi Beta Kappa Honor Society** – 2024
- ★ **Sophia Freund Prize** for the highest ranking undergraduates (*Harvard University*) – 2024
- ★ **Carol Davis Prize** for exceptional service to the physics community (*Harvard University*) – 2024
- ★ **Harvard College Office of Undergraduate Research and Fellowships Conference Funding** – 2023 (twice)
- ★ **Harvard College Research Program Fellowship** – 2022, 2023
- ★ **John Harvard Scholarship** for students in the top 5% of their class (*Harvard University*) – 2021, 2022, 2023
- ★ **Detur Prize** for sophomores with very high academic standing (*Harvard University*) – 2022
- ★ **Program for Research in Science and Engineering Fellowship** (*Harvard University*) – 2021

Research Experience

- ★ **Thesis Research** (Department of Astronomy, Yale University)
 - *Advisor:* Priyamvada Natarajan
 - Beginning work on my thesis, which will focus on massive black holes that wander off-center in their host galaxies due to minor galaxy mergers
- ★ **Second-Year Theory Project** (Department of Astronomy, Yale University)
 - *Advisor:* Priyamvada Natarajan
 - Used the cosmological simulations ASTRID and IllustrisTNG to investigate how the evolutionary histories of massive black holes impact their growth relative to their host galaxies
- ★ **First-Year Observational Project** (Department of Astronomy, Yale University)
 - *Advisor:* Pieter van Dokkum
 - Used Hubble imaging to study the orientation of radio jets relative to nuclear dust features in early-type galaxies

★ **Undergraduate Research** (Center for Astrophysics | Harvard & Smithsonian)

- *Advisor:* Fabio Pacucci
- Used the cosmological simulations ASTRID and IllustrisTNG to study wandering intermediate-mass black holes, over-massive central black holes, and high-redshift massive galaxy quenching

Publications

7. **Weller E. J.**, Natarajan P., Burke C. J., *The Black Hole Occupation Fraction as a Fossil Record of Seeding, Dynamics, and Galaxy Assembly*, 2026, ApJL, 1008, L54, [10.3847/2041-8213/ae9c38](https://arxiv.org/abs/10.3847/2041-8213/ae9c38)
6. **Weller E. J.**, Natarajan P., Burke C. J., Dattathri S., *Tracing Black Hole and Galaxy Growth Across Environments Since Cosmic Noon*, 2026, ApJ, 1008, 70, [10.3847/1538-4357/ae9288](https://arxiv.org/abs/10.3847/1538-4357/ae9288)
5. **Weller E. J.**, van Dokkum P., *The Perpendicularity of Dust Lanes and Radio Jets in Early-type Galaxies: Implications for Active Galactic Nuclei Feedback*, 2026, ApJ, 999, 33, [10.3847/1538-4357/ae3dda](https://arxiv.org/abs/10.3847/1538-4357/ae3dda)
4. **Weller E. J.**, Pacucci F., Ni Y., Hernquist L., Park M., *Discrepancies between JWST Observations and Simulations of Quenched Massive Galaxies at $z > 3$: A Comparative Study with IllustrisTNG and ASTRID*, 2025, ApJ, 979, 181, [10.3847/1538-4357/ada360](https://arxiv.org/abs/10.3847/1538-4357/ada360)
3. **Weller E. J.**, Pacucci F., Natarajan P., Di Matteo T., *Over-massive Central Black Holes in the Cosmological Simulations ASTRID and Illustris TNG50*, 2023, MNRAS, 522, 4963, [10.1093/mnras/stad1362](https://arxiv.org/abs/10.1093/mnras/stad1362)
2. **Weller E. J.**, Pacucci F., Ni Y., Chen N., Di Matteo T., Siwek M., Hernquist L., *Orbital and Radiative Properties of Wandering Intermediate-Mass Black Holes in the ASTRID Simulation*, 2023, MNRAS, 520, 3955, [10.1093/mnras/stad347](https://arxiv.org/abs/10.1093/mnras/stad347)
1. **Weller E. J.**, Pacucci F., Hernquist L., Bose S., *Dynamics of Intermediate-Mass Black Holes Wandering in the Milky Way Galaxy Using the Illustris TNG50 Simulation*, 2022, MNRAS, 511, 2229, [10.1093/mnras/stac179](https://arxiv.org/abs/10.1093/mnras/stac179)

Presentations

- ★ **Massive Black Holes Across Cosmic Time** (*Kavli Institute for Cosmology, Cambridge*) – Sept. 2025
- ★ **Prospects in Theoretical Physics** (*Institute for Advanced Study*) – July 2025
- ★ **Astronomy Club Guest Speaker** (*Mount Si High School; virtual*) – Jan. 2025
- ★ **243rd Meeting of the American Astronomical Society** (*New Orleans, LA*) – Jan. 2024
- ★ **Boston-Area Black Hole Accretion Meeting** (*Harvard University*) – Oct. 2023
- ★ **Harvard-Heidelberg Star Formation Workshop** (*Harvard University*) – Oct. 2023
- ★ **242nd Meeting of the American Astronomical Society** (*Albuquerque, NM*) – June 2023
- ★ **241st Meeting of the American Astronomical Society** (*Seattle, WA*) – Jan. 2023
- ★ **240th Meeting of the American Astronomical Society** (*Pasadena, CA*) – June 2022
- ★ **Undergraduate Research Spotlight** (*Harvard University*) – Mar. 2022
- ★ **Black Holes Across Space and Time Workshop** (*Black Hole Initiative; virtual*) – Dec. 2021

Press Coverage

- ★ [Astrobites article](#) about my paper – Aug. 2026

Teaching Experience

- ★ Teaching Fellow – Astronomy 1800: Introduction to Relativity and Black Holes – Yale University – Fall 2025
- ★ Teaching Fellow – Astronomy 220: Galaxies and Cosmology – Yale University – Spring 2025
- ★ Teaching Fellow – Astronomy 110: Planets and Stars – Yale University – Fall 2024
- ★ Course Assistant – Astronomy 16: Stellar and Planetary Astronomy – Harvard University – Spring 2023, 2024
- ★ Coach – Wintersession Python Introduction – Harvard University – January 2021

Mentoring & Community

- ★ Mentor and Organizing Committee Member – Yale First-Year Astronomy Buddy Program – Summer 2025 - present
- ★ Mentor for undergraduate astronomy students – Yale AstroSibs Program – Fall 2024 - present
- ★ Member – Yale Astronomy Climate and Diversity Committee – Fall 2024 - present
- ★ Co-advisor for high school student – Choate Rosemary Hall Science Research Program – Summer 2025
- ★ Assistant Instructor – Yale Astronomy Research Preparation Series – Spring 2025
- ★ Mentor for beginning undergraduate astronomy students – Harvard AstroSoc – Fall 2023 - Spring 2024
- ★ Events and Panels Chair – Harvard Society of Physics Students – Fall 2021 - Spring 2024